

Sales Analysis

1) Project Overview:

This project focuses on analysing sales data to derive meaningful business insights and support data-driven decision-making. The dataset contains detailed transactional information, including product details, sales performance, customer behaviour, pricing, discounts, and payment methods.

2) Dataset Summary:

Rows: 1000

Columns: 14

- **Product Info:** product_id, product category
- **Sales Details:** sale date, sales amount, quantity sold
- **Pricing:** unit cost, unit price, discount
- **Sales Performance:** sales rep, region, region and sales rep
- **Customer Info:** customer type
- **Transaction Info:** payment method, sales channel

3) Exploratory Data Analysis using Python:

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure and `df.describe()` for summary statistics.

	Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Category	Unit_Cost	Unit_Price	Customer_Type	Discount	Payment_Method
0	1052	03-02-2023	Bob	North	5053.97	18	Furniture	152.75	267.22	Returning	0.09	Cash
1	1093	21-04-2023	Bob	West	4384.02	17	Furniture	3816.39	4209.44	Returning	0.11	Cash
2	1015	21-09-2023	David	South	4631.23	30	Food	261.56	371.40	Returning	0.20	Bank Transfer
3	1072	24-08-2023	Bob	South	2167.94	39	Clothing	4330.03	4467.75	New	0.02	Credit Card
4	1061	24-03-2023	Charlie	East	3750.20	13	Electronics	637.37	692.71	New	0.08	Credit Card
...	...	...	...	...	...	...	...	...	...	...	...	...
995	1010	15-04-2023	Charlie	North	4733.88	4	Food	4943.03	5442.15	Returning	0.29	Cash
996	1067	07-09-2023	Bob	North	4716.36	37	Clothing	1754.32	1856.40	New	0.21	Bank Transfer
997	1018	27-04-2023	David	South	7629.70	17	Clothing	355.72	438.27	Returning	0.06	Bank Transfer
998	1100	20-12-2023	David	West	1629.47	39	Electronics	3685.03	3743.39	New	0.01	Bank Transfer
999	1086	16-08-2023	Alice	East	4923.93	48	Food	2632.58	2926.68	Returning	0.14	Cash

1000 rows × 14 columns

Sales_Channel	Region_and_Sales_Rep
Online	North-Bob
Retail	West-Bob
Retail	South-David
Retail	South-Bob
Online	East-Charlie
...	...
Online	North-Charlie
Retail	North-Bob
Online	South-David
Online	West-David
Online	East-Alice

- **Missing Data Handling:** Checked for null values
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Database Integration:** Connected Python script to “SQL Server” and loaded the cleaned Data Frame into the database for SQL analysis.

4) Data Analysis Using SQL:

We performed structured analysis in SQL Server.

1. Total Sales By Region:

	region	total_sales
1	East	1259792.93
2	North	1369612.51
3	South	1154250.86
4	West	1235608.93

2. Top 5 Products by Sales:

	product_id	total_sales
1	1099	101773.87
2	1092	90615.62
3	1033	89130.41
4	1090	88043.25
5	1086	82269.71

3. Sales by Each Sales Representative:

	sales_rep	total_sales
1	David	1141737.36
2	Bob	1080990.63
3	Eve	970183.99
4	Alice	965541.77
5	Charlie	860811.48

4. Average Discount by Customer Type:

	customer_type	avg_discount
1	New	0.15172619047619
2	Returning	0.153064516129032

5. Monthly Sales Trend:

	year	month	total_sales
1	2023	1	150983.24
2	2023	2	145200.18
3	2023	3	179857.17
4	2023	4	141896.8
5	2023	5	133039.23
6	2023	6	115051.25
7	2023	7	170739.06
8	2023	8	183617.33
9	2023	9	149510.72
10	2023	10	142976.82
11	2023	11	160565.59
12	2023	12	187974.98
13	2024	1	19328.01

6. Most Sold Product Category:

	product_category	total_qty
1	Clothing	6922
2	Furniture	6729
3	Electronics	6096
4	Food	5608

7. Profit Calculation:

	product_id	unit_price	unit_cost	quantity_sold	profit
1	1008	3385.61	2907.22	49	23441.11
2	1090	5253.07	4766.53	48	23353.92
3	1066	1805.56	1319.24	47	22857.04
4	1012	803.71	325.75	47	22464.12
5	1035	4624.1	4132.79	45	22108.95
6	1060	4276.99	3784.96	44	21649.32
7	1061	1844.8	1364.51	45	21613.05
8	1014	2747.81	2257.48	44	21574.52
9	1050	586.18	112.35	45	21322.35
10	1081	1844.07	1362.15	44	21204.48
11	1001	5316.13	4865.33	47	21187.6
12	1063	1669.95	1170.07	42	20994.96
13	1053	2943.31	2443.57	42	20989.08
14	1062	1462.19	987.74	44	20875.8
15	1058	3049.04	2614.48	48	20858.88
16	1074	2611.94	2120.81	42	20627.46

8. High Value Transactions:

	sales_amount	discount
1	9744.52	0.09
2	9582.12	0.01
3	9914.15	0.02
4	9183.11	0.02
5	9088.2	0.09
6	9413.77	0.09
7	9733.47	0.01
8	9671.77	0.08
9	9450.04	0.05
10	9805.65	0.04
11	9688.52	0.03
12	9653.65	0.05
13	9472.66	0.05
14	9215.59	0.04
15	9146.51	0.07
16	9602.64	0.08
17	9027.56	0.06
18	9989.04	0.05
19	9298.48	0.04
20	9813.07	0.03
21	9733.46	0.09
22	9680.84	0.07
23	9702.27	0.01

9. Payment Method Analysis:

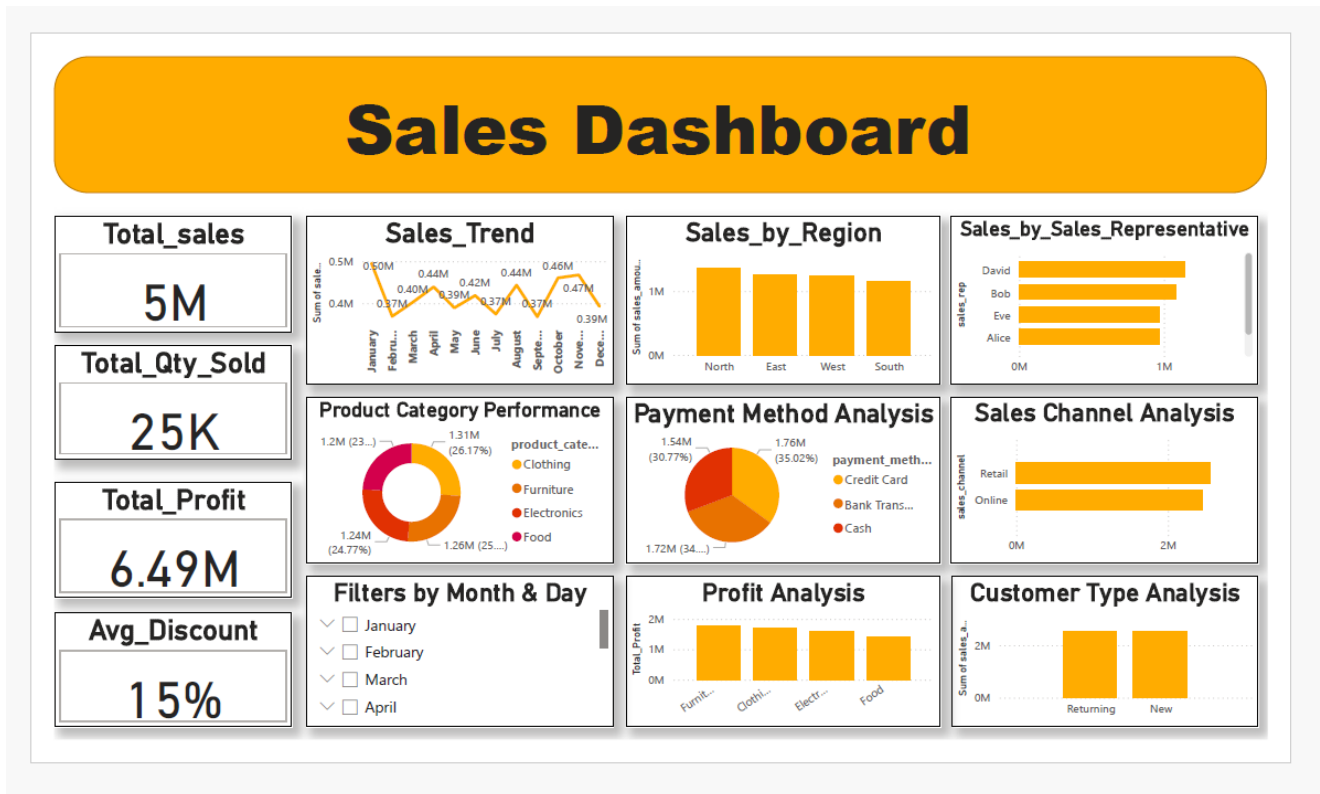
	payment_method	total_sales
1	Bank Transfer	1717363.17
2	Cash	1544338.54
3	Credit Card	1757563.52

10. Best Performing Region & Sales Rep Combination:

	region	sales_rep	total_sales
1	South	David	311416.99
2	East	Bob	309876.11
3	North	Eve	304172.2
4	North	Charlie	299789.01
5	West	Bob	286449.28
6	East	David	284613.51
7	North	David	282539.27
8	West	David	263167.59
9	East	Alice	260789.47
10	South	Bob	251200.43
11	North	Alice	249647.22
12	West	Charlie	239708.85
13	South	Alice	239414.54
14	East	Eve	239340.9
15	North	Bob	233464.81
16	West	Eve	230592.67
17	West	Alice	215690.54
18	South	Eve	196078.22
19	East	Charlie	165172.94
20	South	Charlie	156140.68

5) Power Bi Dashboard:

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6) Business Recommendation:

1. Focus on High-Performing Regions:

From **“Total Sales by Region”** analysis

- Invest more marketing budget and inventory in top-performing regions
- Expand operations in these areas

2. Improve Low-Performing Regions:

- Identify regions with low sales
- Provide targeted discounts, local campaigns, or better distribution

3. Optimize Sales Representative Performance:

From **“Sales by Each Sales Representative”**

- Reward top performers with incentives
- Train low-performing reps

4. Control Discount Strategy:

From **“Average Discount by Customer Type”**

- Avoid giving high discounts where not needed
- Offer discounts only where they increase sales volume

5. Promote High-Profit Products:

From **“Profit Calculation”**

- Identify high-margin products
- Promote and bundle them

6. Optimize Sales Channels & Payment Methods:

From **“Payment Method Analysis” & “Sales Channel”**

- Focus on top-performing channels (Online/Offline)
- Promote preferred payment methods

